2	A student constructs a potentiometer as shown in Fig. 2.1. The driver battery has negligible internal resistance and an e.m.f. of E_1 = 2.0 V. Wire PQ has a total length of 50.0 cm and resistivity ρ. Wire PQ also has a total resistance of 10.0 Ω. Unfortunately, there is a defect in wire PQ (shaded in Fig. 2.1) where that part of the wire has a cross-sectional area, S, that is half of the rest of the wire. This defective portion has a length that is denoted by x.

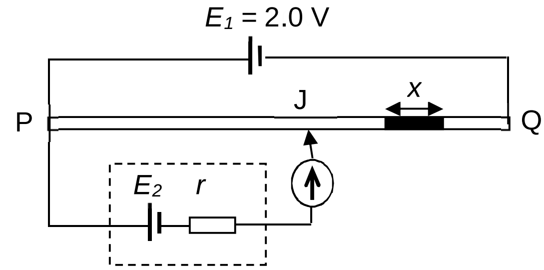

Fig. 2.1

The student carried out an experiment and determined that the e.m.f. E_2 was 1.01 V. The balance length PJ which did not include the defective portion, was 25.5 cm.

(a) When the potentiometer is balanced, determine the current I_{PQ} in the wire PQ.

current, I_{PQ} = ... A [1]

$$I = \frac{V}{R} = \frac{2.0}{10.0} = 0.20 \text{ A}$$

(b) **(i)** Using your answer from **(a)**, obtain an expression for

1. V_{PJ}, the potential difference between points P and J, in terms of S and ρ;

[1]

$$V_{PJ} = I_{PQ} R_{PJ}$$

$$= (0.20)\ \frac{\rho(0.255)}{2S}$$

$$= 0.0255\ \frac{\rho}{S}$$

Accept if candidate did not convert 25.5 cm to 0.255 m.

2. V_{PQ}, the potential difference between points P and Q, in terms of ρ, S and x.

[1]

$$V_{PQ} = I_{PQ} R_{PQ}$$

$$= (0.20)\left[\frac{\rho(0.500 - x)}{2S} + \frac{\rho x}{S}\right]$$

Accept if candidate did not convert 50.0 cm to 0.500 m.

3. Hence determine the defective length, x.

		x =	.... cm	[2]

From part 1: $V_{PJ} = 0.0255\dfrac{\rho}{S}$ ---------------------- (1)

From part 2: $V_{PQ} = (0.20)\left[\dfrac{\rho(0.50-x)}{2S} + \dfrac{\rho x}{S}\right]$-----------(2)

$\dfrac{(1)}{(2)}$: $\dfrac{V_{PJ}}{V_{PQ}} = \dfrac{0.0255\dfrac{\rho}{S}}{(0.20)\left[\dfrac{\rho(0.50-x)}{2S} + \dfrac{\rho x}{S}\right]}$

$\dfrac{1.01}{2.0} = 0.5\left[\dfrac{1}{(0.50-x)+2x}\right]$

$0.50495 = 0.50 + x$

$x = 4.95 \times 10^{-3}$ m = 0.495 cm

(c) In a further experiment, the student swaps the connections to the wire PQ and repeats the measurement, as shown in Fig. 2.2.

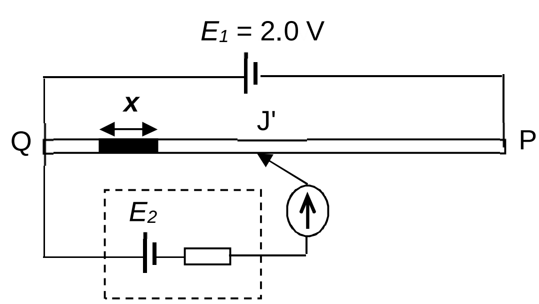

Fig. 2.2

State and explain if the new balance length QJ' would be shorter, longer or unchanged from the set-up in Fig. 2.1.

....

....

....

....

....

.... [4]

For balance length, <u>the potential difference across QJ' must still be equal to E_2</u>.

<u>Since the same current flow through PQ, the resistance of length QJ' of the potentiometer wire must be the same/equal as before</u>.

		Since the <u>defective portion has a smaller cross-sectional area, it would have a higher resistance compared to the non-defective portion of similar length.</u> **the balance length QJ' is <u>shorter</u>.**	
3	A binary star system shown in Fig.3.1 consists of two stars orbiting around their common centre of mass, B. One such binary star system is known as Sirius where the mass of Sirius		